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Executive Summary

This report is aimed at Real Estate agencies to provide a detailed analysis of Madrid's rental market, with the goal of identifying gaps in the market to help agencies capture opportunities. The report uses data from Idealista.com which consists of 2,089 properties across 20 districts. There are 2 key parts to the analysis conducted: Part 1 uses K-means to segment properties into four distinct clusters, while Part 2 explores a linear regression model on a selected cluster to predict rent prices and evaluate pricing mismatches.

Our analysis concludes that rent prices in Madrid were primarily driven by a few property characteristics, namely size, location and floor. Premium districts identified are Salamanca, Centro and Retiro, with the ability to command higher rental prices. Properties of larger size and located at higher levels also exhibited higher prices. Deeper analysis revealed a significant pricing gap: ~25% of properties are underpriced, while ~20% are overpriced. In particular, the districts of Retiro, Centro and Moncloa had the highest concentration of underpriced properties, signalling clear opportunities for more revenue upside.

Recommendations and actionable steps can be summarised in these 3 points:

1. Capitalise underpriced opportunities in Retiro, Centro and Moncloa
2. Re-evaluate overpriced properties in Arganzuela, Puente Vallecas and Ciudad Lineal
3. Prioritise onboarding landlords with larger properties located in prime districts

1. Data Audit, Data Cleaning and Data Manipulation

Original data comprised of 2,089 properties listing, with each described by 15 features that can be grouped into four broad categories – (1) Location: District, Area, Address; (2) Structural: Sq.Mt [1], Bedrooms, Floor; (3) Add-ons: Elevator, Outer; (4) Property Type: Penthouse, Cottage, Duplex, Semidetached, as well as the target variable – Rent (in Euros). At a preliminary glance, majority of listings were located at Salamanca (15.0%) and Centro (13.4%). Initial Exploratory Data Analysis (EDA) showed obvious outliers which were then removed to protect data quality *(refer to Part A1.1 in Annex A).*

A binary 'Studio' variable was created to capture apartments with no bedrooms but labelled as *'estudio'*, to prevent any unintended removal. Another round of data cleaning was then conducted to exclude non-essential fields that do not influence rental prices – 'Id', 'Address', 'Number'. Additionally, only 2.4% of listings contained incomplete information on property characteristics. Given the small proportion and the risk of introducing misleading information through statistical imputation, these listings were removed to ensure data cleanliness and maintain data reliability to support the eventual pricing insights *(refer to Part A1.2 in Annex A).* The cleaned dataset contained 1,893 listings.

2. Segmentation

To enable more targeted and actionable pricing recommendations, segmentation analysis was conducted to identify distinct, but internally homogeneous groups of rental properties in Madrid. As the data mainly consisted of numerical and binary variables, K-means clustering was used to perform segmentation, where clusters were determined by minimising within-cluster variations and maximising between-cluster variation.

[1] Sq.Mt refers to Square-metre, measuring the size of property

2.1 Preparation for K-means Clustering

Seven features were first selected for segmentation based on both current understanding of Madrid rental market and the characteristics that typical influence rent prices: Sq.Mt, Bedrooms, Penthouse, Cottage, Duplex, Semidetached and Studio. These variables reflected property size, capacity, and housing format, which were inherent characteristics of the property and less susceptible to external demand forces such as the location.

Correlation analysis *(refer to Part B2.2 in Annex B)* was conducted to assess potential redundancy among selected features. 2 pairs showed notable correlation: Cottage and Semidetached (0.82) likely reflected overlapping housing classifications, while Bedrooms and Sq.Mt (0.76) are both related to property size and capacity, where larger properties tend to have more bedrooms. Despite these correlations, all features were retained given that each captured meaningful structural variation. These features were then rescaled to ensure that no single variable dominated the grouping process – for example, Sq.Mt had a high mean of 115.7 and thus imposed greater influence without rescaling (*refer to Part B2.3 in Annex B).*

2.2 Determining the Optimal Number of Clusters

Both Elbow Method and Silhouette Analysis pointed to the same conclusion of 4 clusters as the optimal number, with the highest silhouette score of 0.668 at k=4 *(full technical explanation can be found in Parts B2.4 – B2.5 in Annex B).*

2.3 Feature Importance & Validation

Property type indicators – 'Duplex', 'Semidetached' and 'Studio' were then identified as the primary drivers of segmentation with each having a maximum relative importance of 1.00. ANOVA test was also conducted on the seven features, which further validated that the segmentation effectively captured distinct property groups (*refer to Parts B2.6 – B2.7 in Annex B for detailed statistical tests and analysis).*

2.4 Final Clusters

Final Clusters are as follow – Cluster 1: Studio Apartments, Cluster 2: Mid-Size Apartments, Cluster 3: Large Duplex Units and Cluster 4: Luxury Detached Houses *(Refer to Parts B2.8 – B2.10 in Annex B for detailed labelling and profiling characteristics).*

<u>3. Linear Regression</u>

The Linear Regression Modelling and Analysis was evaluated on Cluster 2: Mid-Size Apartments, given that it is the largest cluster and thus most representative of Madrid's rental market. Regression modelling was performed to first determine the most appropriate model to predict rent prices, in order to facilitate meaningful insights and action steps which are addressed in Part 4. The detailed technical explanation and breakdown of arriving at the final linear regression model for predicting rent prices can be found in Annex C.

3.1 Overview of Selected Group – Cluster 2

Current rent prices for this selected group averaged at €1,848, with most listings having a price of €1,400. Most properties are also located within the Salamanca (16.4%) and Centro (13.0%) districts, while the size of these properties ranges from 20m^2 to 500m^2.

3.2 Technical Preparation Before Modelling

(Refer to Parts C3.2 to C3.4 in Annex C for full run-through of technical approach)

Before the linear regression model can be evaluated on the datasets, several technical steps were done to prepare the setup adequately for modelling. These included validation, removal of redundancy variables and finally selecting the most applicable explanatory variables for the regression model.

In preparing the data for modelling, certain variables were excluded from the eventual model: 'Cluster'' and 'Area' were excluded as they did not have any analytical values; 'Duplex', 'Semidetached' and 'Studio' were dropped as constant variables that no longer drove pricing differences; 'Bedrooms' (VIF = 12.88) and 'Elevator' (VIF = 10.33) were removed due to multicollinearity, which is aligned with the strong correlations with 'Sq.Mt' and 'Floor'.

Statistical significance via Backward elimination was examined next to ensure that only statistically significant variables are retained in the model. The features retained spanned property size, building characteristics: Sq.Mt, Floor, Cottage, and seven district indicators (Centro, Chamartín, Chamberí, Moncloa, Retiro, Salamanca, and Tetuán). The resulting model containing these 10 final features produced an R-squared of 0.71, indicating that the model wasable to explain 71% of the variances.

Rescaling the final features, it was evident that property size (Sq.Mt) was the strongest driver of rent prices (standardised beta = 989.94. The district location was the next most important driver of prices – Salamanca (320.66), Centro (264.69), Chamberi (164.85) and Chamartin (139.43) command the largest district premiums. Floor level also influenced prices upwards (83.65) – higher-floor properties have higher rents. The negative coefficient for Cottage was likely attributed to its limited presence in the data.

3.3 Selecting the Type of Model: Level-Level vs Log-Level

Level-Level regression model was first estimated on the Train dataset which produced an R-squared of 0.71. This confirmed that the model effectively captured structural and locational drivers of rent. However, rent prices were earlier identified as right skewed and residual diagnostics indicated presence of slight heteroskedasticity *(Refer to Part C3.5 in Annex C)*. A Log-Level specification was therefore tested to explore different types of model.

While log transformation helped to improve residual behaviour – increased normality and reduced heteroskedasticity, the back-transformation required for interpretability in Euros introduced additional distortion, especially at higher prices. This led to reduced model performance, with R-squared declining to 0.56. Residual diagnostics also worsened, with the Root Mean Square Error (RMSE) increasing from €703 to €868 *(Refer to Part C3.6 in Annex C)*. Considering the results of the Log-Level model and the nature of the rental markets, the Level-Level model was thus selected due to better predictive accuracy.

3.4 Evaluating Predictive Performance and Final Linear Regression Model

The selected Level-Level model was then applied sequentially on the Test data *(refer to Part C3.7 in Annex C)* followed by the Reserved data, with the latter being used to confirm the model's out of sample robustness.

The predictive performance of the chosen Level-Level model was stable across all three datasets – specifically, R-squared is consistently around 0.70 *(refer to Part C3.8 in Annex C)*, indicating that the model was able to effectively capture and explain 70% of the variance. The error metrics likewise demonstrated minimal differences across, thereby further confirming the consistency and thus stability of the model. In conclusion, the Level-Level Linear Regression Model generalised well and did not exhibit signs of overfitting.

4. Business Application: Pricing Analysis and Recommendations

The validated pricing model was applied across all 1,741 Mid-size Apartments listings to benchmark each property's listed prices against the model-predicted prices – 22% were underpriced, 19% were overpriced while the remaining were fairly priced within a +/- 20% band. Overall, although the average pricing gap is +1.97%, the median gap of -5.69% suggested that typical listings were slightly underpriced compared to the structural fundamentals.

Summary of Key Insights: Retiro, Centro, and Moncloa have the highest share of underpriced properties (34.5–36.8%), while Arganzuela, Ciudad Lineal and Puente Vallecas have the highest proportion of overpriced properties (~40-60%). Property size was the primary driver of rental value across the market, while Salamanca, Centro, Chamberi and Chamartin had the largest location premiums. Lastly, higher floors also commanded higher rents overall *(refer to Annex D for full details on insights)*.

Key Recommendations:

1) Capitalise on underpriced opportunities in Retiro, Centro and Moncloa

Opportunities arising from underpriced properties were found mainly in these districts and represents significant missed revenue opportunities for landlords. Real estate agencies should reach out to landlords within these districts to advise upward price adjustments.

2) Re-evaluate overpriced properties in Arganzuela, Puente Vallecas, and Ciudad Lineal

These districts have a disproportionately higher proportion of overpriced properties, with Arganzuela having the highest at 60%. More specifically, the property with the largest pricing gap is in Arganzuela with a pricing gap of +502.82% (see Figures D2 and D4). Due to the huge mismatch in prices and therefore expensive rentals, it is possible that these listings face prolonged periods of vacancies. Real estate agencies should work with landlords to adjust prices downwards, to reduce the occurrence of missed revenue opportunity windows.

3) Prioritise landlords with larger properties in prime districts

Earlier analysis concluded that property size and locations are the strongest drivers of rent prices in Madrid. In fact, amongst the top 10 underpriced properties, 70% are in identified premium regions (Salamanca and Centro). Furthermore, properties in prime locations are effectively Cash Cows in the rental market, given the more consistent demand and ability to command higher prices. As such, real estate agencies should prioritise landlords with larger properties in prime districts to benefit from the huge upside available.

Note: These pricing insights and recommendations are limited to mid-sized apartments (i.e. Cluster 2). Other types of apartments such as studios, luxury properties will require a separate evaluative framework to identify key factors affecting rent prices, and therefore recommendations.

TECHNICAL ANNEX A – DATA MANIPULATION

This annex supports Section 1 of the main report and aligns with the technical Python notebook.

A1.1 Data Exploration and Data Cleaning

Initial exploratory analysis of the data revealed several data quality issues.

	Id	Rent	Bedrooms	Sq.Mt	Floor	Outer	Elevator	Penthouse	Cottage	Duplex	Semidetached
count	2089.00	2089.00	2000.00	2089.00	1948.00	1927.00	1956.00	2089.00	2089.00	2089.00	2089.00
mean	1094.03	1932.25	2.48	128.92	25.66	0.87	0.88	0.08	0.04	0.03	0.01
std	630.61	1495.47	1.31	115.75	975.07	0.34	0.32	0.27	0.20	0.17	0.12
min	1.00	450.00	0.00	15.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00
25%	550.00	950.00	2.00	65.00	2.00	1.00	1.00	0.00	0.00	0.00	0.00
50%	1094.00	1400.00	2.00	90.00	3.00	1.00	1.00	0.00	0.00	0.00	0.00
75%	1636.00	2500.00	3.00	147.00	5.00	1.00	1.00	0.00	0.00	0.00	0.00
max	2188.00	16000.00	8.00	1250.00	43039.00	1.00	1.00	1.00	1.00	1.00	1.00

Figure A1: Summary of Descriptive Statistics of Original Data

The descriptive statistics (Figure A1) revealed the presence of extreme outliers which would distort the modelling if left untreated. The maximum Floor value of 43,039 was unrealistic, impossible and clearly a data entry error. The largest property size of 1,250m^2 was more consistent with commercial real estate properties instead of residential apartments.

Thresholds were established based on current Madrid residential market context – buildings rarely exceeded 50 levels and apartments rarely exceeded 500m^2. Outliers identified using these thresholds were removed and accounted for 7.13% of the entire data, ensuring minimal data loss while protecting data quality.

A1.2 Feature Engineering & Further Data Cleaning

As mentioned in the main report, a binary 'Studio' variable was created – this was done through parsing the 'Address' fields for the keyword *"estudio"*. Originally, these properties had empty bedroom numbers and were subsequently assigned with a bedroom count of 0 after the creation of the new 'Studio' variable. A total of 88 studio apartments were detected.

	dtype	missing_n	missing_%	n_unique
Number	str	1342	64.24	137
Outer	float64	162	7.75	2
Floor	float64	141	6.75	29
Elevator	float64	133	6.37	2
Bedrooms	float64	89	4.26	9
Area	str	4	0.19	140
Id	int64	0	0.00	2089
District	str	0	0.00	20
Address	str	0	0.00	1336
Rent	int64	0	0.00	250
Sq.Mt	int64	0	0.00	276
Penthouse	int64	0	0.00	2
Cottage	int64	0	0.00	2
Duplex	int64	0	0.00	2
Semidetached	int64	0	0.00	2

Figure A2: Summary of Missing and Unique values in Original Data

Three columns were dropped next:

1. 'ID': Served as an identifier, but had no analytical value
2. 'Address': A granular text field; however location information was sufficiently captured in the 'District' variable
3. 'Number': Contained 1,342 missing values (64.24%), making it unsuitable for modelling

The remaining missing values in the 'Outer', 'Floor', 'Elevator' and 'Bedrooms' columns were handled by removing the affected rows entirely to prevent incomplete data issues. Imputing the mean or median was not adopted to prevent introducing misleading data and since these observations only accounted for 2.42%, removing them ensured limited data loss while improving overall data cleanliness.

Final cleaned data to prepare for Segmentation consisted of 1,893 listings, as follows:

	Rent	Bedrooms	Sq.Mt	Floor	Outer	Elevator	Penthouse	Cottage	Duplex	Semidetached	Studio
count	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00	1893.00
mean	1820.10	2.26	114.18	3.59	0.87	0.89	0.08	0.00	0.03	0.00	0.05
std	1277.60	1.24	75.22	3.00	0.33	0.31	0.28	0.04	0.18	0.03	0.21
min	450.00	0.00	20.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25%	950.00	1.00	65.00	2.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
50%	1400.00	2.00	90.00	3.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
75%	2300.00	3.00	136.00	5.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
max	16000.00	8.00	500.00	29.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Figure A3: Summary of Descriptive Statistics of Cleaned Data

TECHNICAL ANNEX B – SEGMENTATION

This annex supports Section 2 of the main report and aligns with the technical Python notebook.

K-means clustering was selected as the appropriate segmentation method as it is well suited to the nature of the data (mostly numerical and binary structural variables), and also aligns with the overarching objective of identifying distinct, non-overlapping property types. Prior removal of extreme outliers further strengthened the suitability of adopting this distance-based approach.

'Rent' was identified as the target variable and therefore excluded from the segmentation process to ensure that clusters identified reflect the similarities in the structural characteristics and were not influenced by the rent prices. This also helped prevent any circular reasoning during subsequent regression analysis.

B2.1 Features Selection and Exploratory Data Analysis (Pre-Segmentation)

In addition to the main report, the following exploratory analysis was done to better understand the features selected:

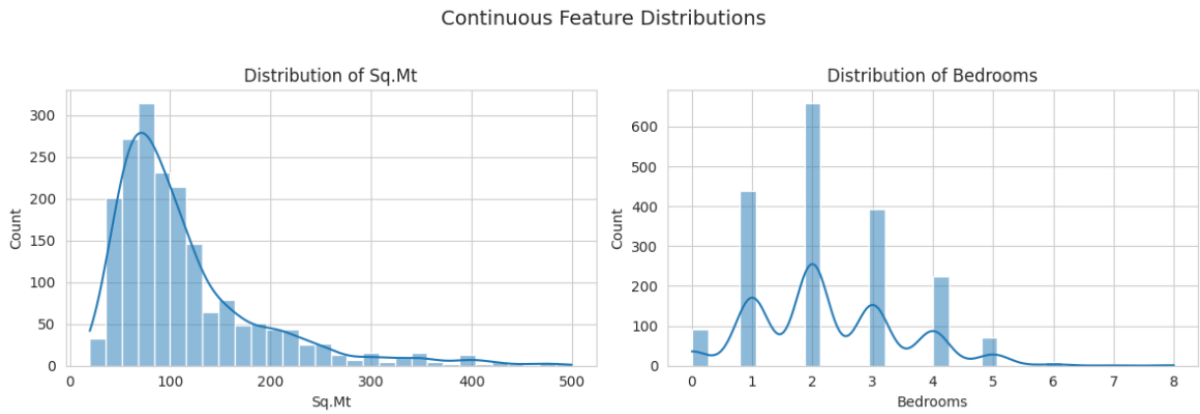


Figure B1: Histogram of Continuous Features

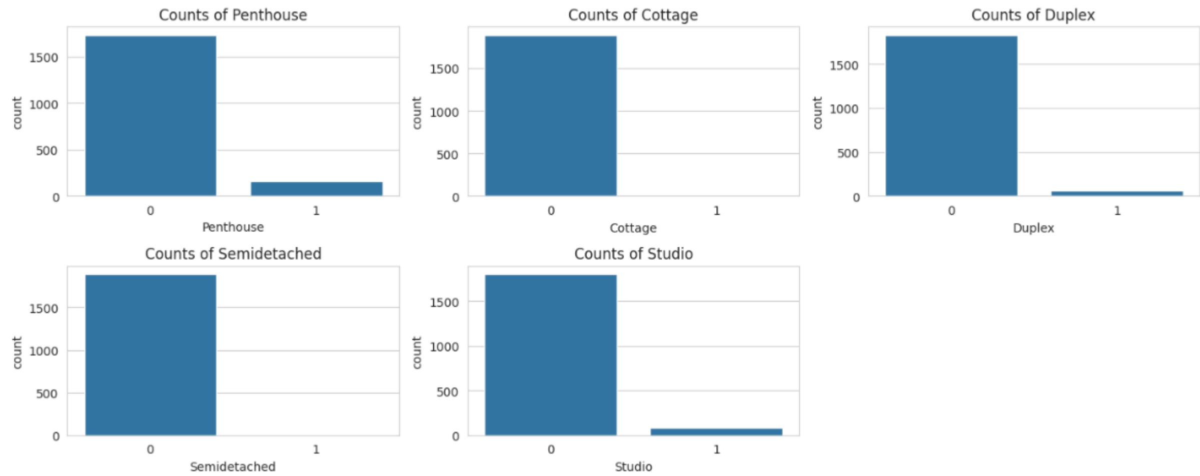


Figure B2: Bar Charts of Binary Features

B2.2 Correlation & Redundancy Analysis

As explained in the main report, the following heatmap was generated to examine the correlations between selected features:

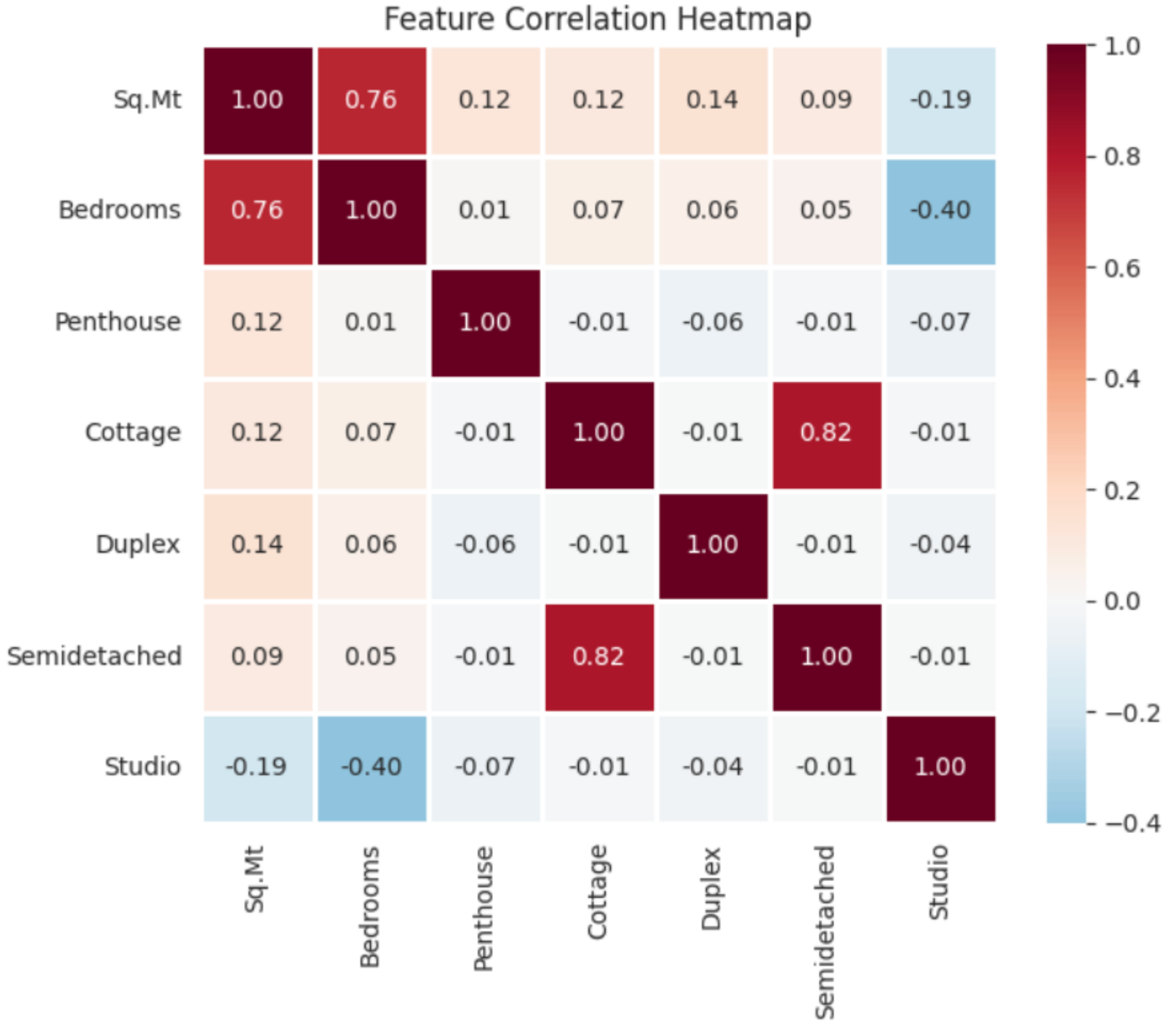


Figure B3: Correlation Heatmap of Selected Features

Aside from the two largest correlation pairs addressed in the main report, all other features show weak correlations, which confirmed that the selected features provided sufficiently distinct information for segmentation.

B2.3 Rescaling

Because K-means clustering looks at the Euclidean distance, features with different numerical scales require standardisation prior to clustering. The following formula was used to calculate the Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{p} (x_{ik} - x_{jk})^2}$$

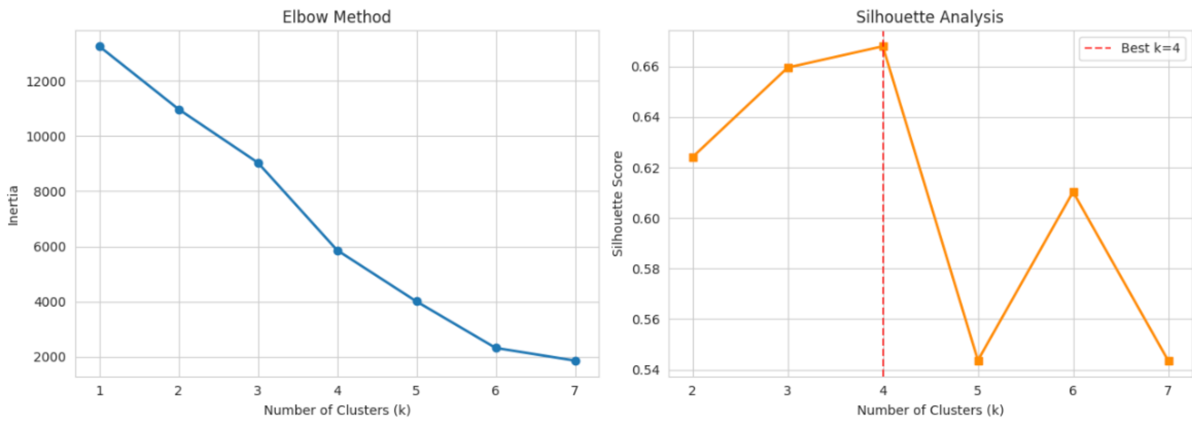
As with the Sq.Mt example identified in the main report, the high mean of 115.7 would mean that without rescaling, property size will dominate the distance computation, and especially when compared to binary features (e.g. Penthouse). Z-score transformation was thus used to standardise and rescale each feature to a mean of 0 and standard deviation of 1, ensuring that all features contribute evenly to the distance metric calculation:

$$Z = \frac{x - \mu}{\sigma}$$

where μ: sample mean and σ: sample standard deviation
Note: Standard Normal Distribution has a mean ≈ 0, standard deviation ≈ 1

The Z-score transformation was preferred over Min-Max scaling because it preserves the relative distributional shape of each variable and was more robust to residual outliers – an important consideration given the heterogeneity of the Madrid rental market.

B2.4 Determining the Optimal Number of Clusters




```
Silhouette scores:
  k=2: 0.624
  k=3: 0.660
  k=4: 0.668 ← best
  k=5: 0.544
  k=6: 0.610
  k=7: 0.544
```

Figure B4: Results of Elbow Method and Silhouette Analysis

The Elbow Method identifies the optimal number of clusters by plotting inertia (sum of intra-cluster distance squared) against the number of clusters. In general, inertia decreases with higher value of k as the observations are grouped more tightly. The optimal value of k was identified at the inflection point where the rate of decline diminishes. As shown in Figure B4, a visible reduction is observed at k = 4-5, suggesting that four clusters are the optimal number.

Silhouette Score measures how well each observation fits within its clusters by evaluating the cohesion (how similar points in each cluster are) and separation (how distinct the clusters are from each other):

$$S(i) = \frac{separation(i) - cohesion(i)}{\max\big(separation(i), cohesion(i)\big)}$$

Silhouette scores range from -1 to 1, where values close to 1 indicate well-defined clusters. As shown in Figure B4, the highest average Silhouette Score is 0.668 at k = 4, which aligned with the conclusion from the Elbow Method analysis, therefore further confirming that k = 4 was the optimal number of clusters.

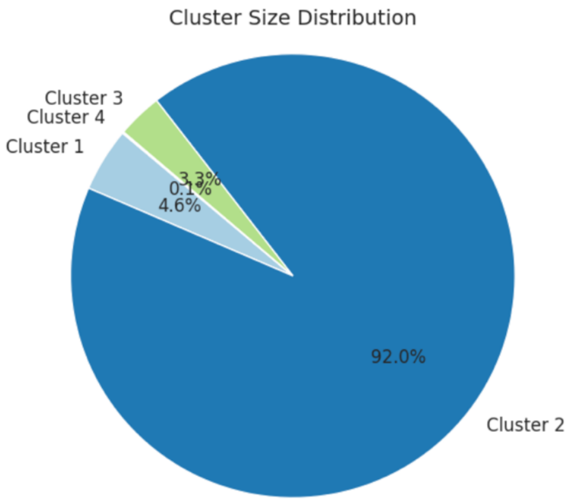


Figure B5: Cluster Size Distribution

Figure B5 illustrates the distribution of properties across the four clusters formed, with Cluster 2 containing a total of 1,741 (92.0%) listings.

B2.5 Global and Individual Silhouette Analysis

Global average silhouette score equates to 0.668, which serves as a reference threshold.

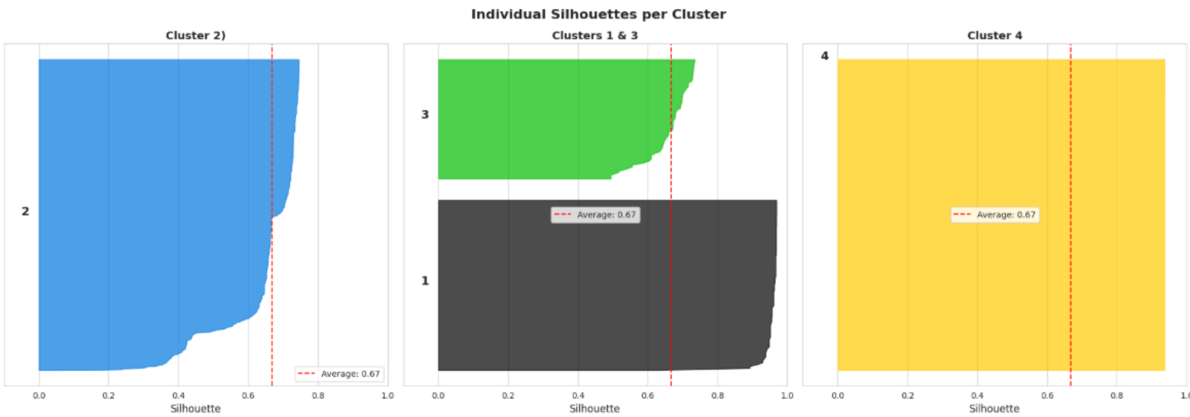


Figure B6: Silhouettes of each Cluster formed

Figure B6 above shows the individual silhouette score per cluster, plotted against the global score of 0.668 (line in red). These individual silhouette plots confirmed the overall robustness of the 4-cluster selection. Clusters 3 and 4 demonstrate strong cohesion with majority of observations scoring above the global average of 0.67. Clusters 1 and 2, while having wider distributions, recorded no negative values which provided assurance that no observations were incorrectly assigned. This wider spread in scores is expected given that both clusters are larger in size.

B2.6 Relative importance of each variable

```
Variable Importances:

              relative importance
Duplex                       1.00
Semidetached                 1.00
Studio                       1.00
Cottage                      0.67
Bedrooms                     0.17
Sq.Mt                        0.06
Penthouse                    0.01
```

Figure B7: Relative Importance of Each Feature

The variable importance measures each feature's relative contribution to the cluster separation. Among these features driving segmentation, property type indicators (Duplex, Semidetached and Studio) had the highest relative importance at 1.00, followed by Cottage (0.67). Continuous variables contributed at lower magnitudes: bedrooms (0.17), square meterage (0.06), and penthouse (0.01).

B2.7 Run ANOVA test

ANOVA tests whether the mean of a variable differs significantly across multiple groups and if they are statistically distinct, with the following hypotheses:

H_0: The mean of the variable is equal across all clusters

H_1: At least one cluster mean differs

Results of the ANOVA test is summarised below:

	Variable	F-statistic	p-value	Overall Mean
0	Sq.Mt	42.25	0.00	114.18
1	Bedrooms	125.02	0.00	2.26
3	Cottage	1257.42	0.00	0.00
4	Duplex	inf	0.00	0.03
5	Semidetached	inf	0.00	0.00
6	Studio	inf	0.00	0.05
2	Penthouse	5.08	0.00	0.08

Figure B8: Results of ANOVA Test

ANOVA F-statistic calculation is as follows:

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

A larger F-statistic indicates strong between-cluster variation relative to within-cluster dispersion. As shown in Figure B8, all features returned p-values of 0.00, confirming that the null hypothesis was rejected for each variable and that each feature differed significantly across clusters.

Diving deeper, 'Duplex', 'Semidetached' and 'Studio' features returned infinite F-statistics which indicated perfect separation across clusters. It can thus be inferred that these features were exclusive to specific clusters and not present across all clusters. Cottage produced extremely high F-statistic of 1,257, further confirming strong between-cluster variation. Bedrooms (125.02) and Sq.Mt (42.25) displayed meaningful separation, while Penthouse (5.08) remained statistically significant albeit having the lowest F-statistic score.

B2.8 Cluster profiles

```
Cluster means (structural features):
```

Cluster	Sq.Mt	Bedrooms	Penthouse	Cottage	Duplex	Semidetached	Studio	Average	Std.dev
1	49.56	0.00	0.00	0.00	0.00	0.00	1.00	7.22	17.29
2	115.10	2.36	0.09	0.00	0.00	0.00	0.00	16.79	40.14
3	173.11	2.66	0.00	0.00	1.00	0.00	0.00	25.25	60.37
4	325.00	4.00	0.00	1.00	0.00	1.00	0.00	47.29	113.38

Figure B9: Cluster Means Table

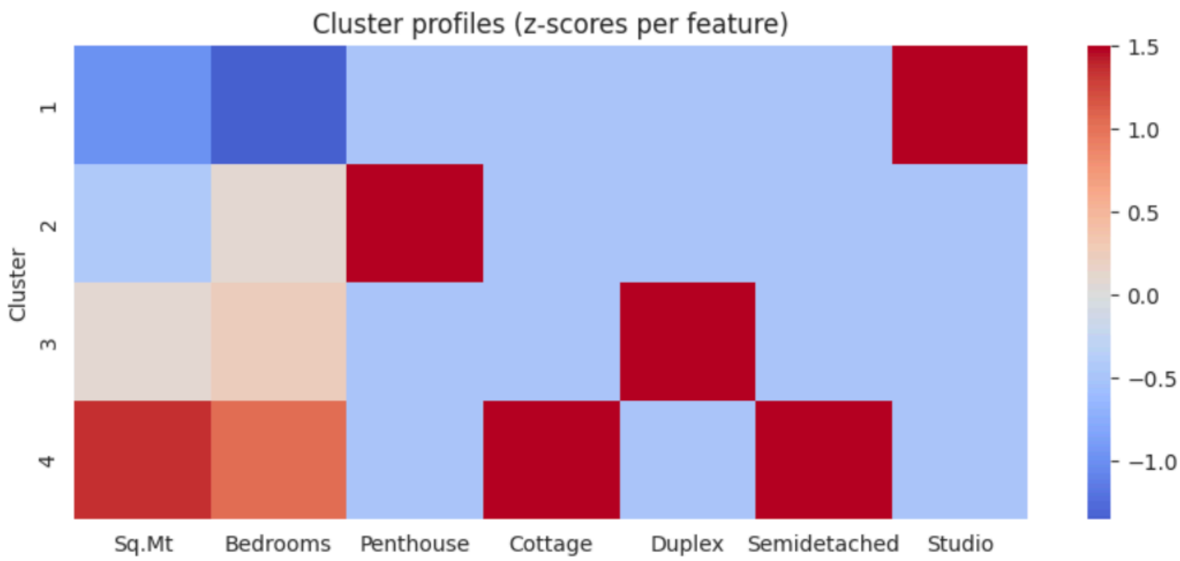


Figure B10: Heatmap of Z-scores by Clusters and Features

Figures B9 and B10 confirmed four structurally distinct property types – Cluster 1 was defined entirely by the Studio indicator (1.00) with the smallest average size of 49.56m²; Cluster 2 consisted of moderate-sized apartments of ~115.10m² with an average of 2.36 bedrooms, but with no special property type indicators; Cluster 3 was clearly identified by the Duplex indicator (1.00) and also had a slightly large average property size of 173.11m² with 2.66 bedrooms; Cluster 4 had an average size of 325m² and 4 bedrooms, with Cottage (1.00) and Semidetached (1.00) indicators, and hence can be safely inferred as large detached properties.

B2.9 Reattach rent for economic profiling

As a final step to prepare for economic profiling, the 'Rent' variable was added back to the dataframe to enable meaningful analysis:

Rent distribution by cluster:

Cluster	count	mean	std	min	25%	50%	75%	max
1	88.00	907.50	281.48	450.00	700.00	850.00	1000.00	2150.00
2	1741.00	1848.45	1283.65	450.00	990.00	1400.00	2400.00	16000.00
3	62.00	2297.39	1423.14	700.00	1212.50	2100.00	3111.75	8500.00
4	2.00	2500.00	989.95	1800.00	2150.00	2500.00	2850.00	3200.00

Figure B11: Rent Distribution by Clusters

B2.10 Determining cluster labels

The respective cluster labels and descriptions are as follow:

Cluster 1 **Studio Apartments**	Studio apartments with an average size of 49.56m^2 and with the lowest average rent of €908.
Cluster 2 **Mid-Size Apartments**	The dominant market segment with an average size of 115.10m^2 and 2.36 bedrooms has an average rent of €1,848.
Cluster 3 **Large Duplex Units**	Multi-level properties with an average size of 173.11m^2 and 2.66 bedrooms, averaging a higher rent of € 2,297.
Cluster 4 **Luxury Detached Houses**	Largest properties in the market with an average size of 325m^2 and 4 bedrooms, averaging the highest rent of €2,500.

TECHNICAL ANNEX C – LINEAR REGRESSION

This annex supports Section 3 of the main report and aligns with the technical Python notebook.

The purpose of the Linear Regression is to predict rent prices. As such, the target and dependent variable in the model is the variable 'Rent'.

To identify the most appropriate and optimal Linear Regression Model for rent prices prediction, the following diagram illustrates our overall technical approach:

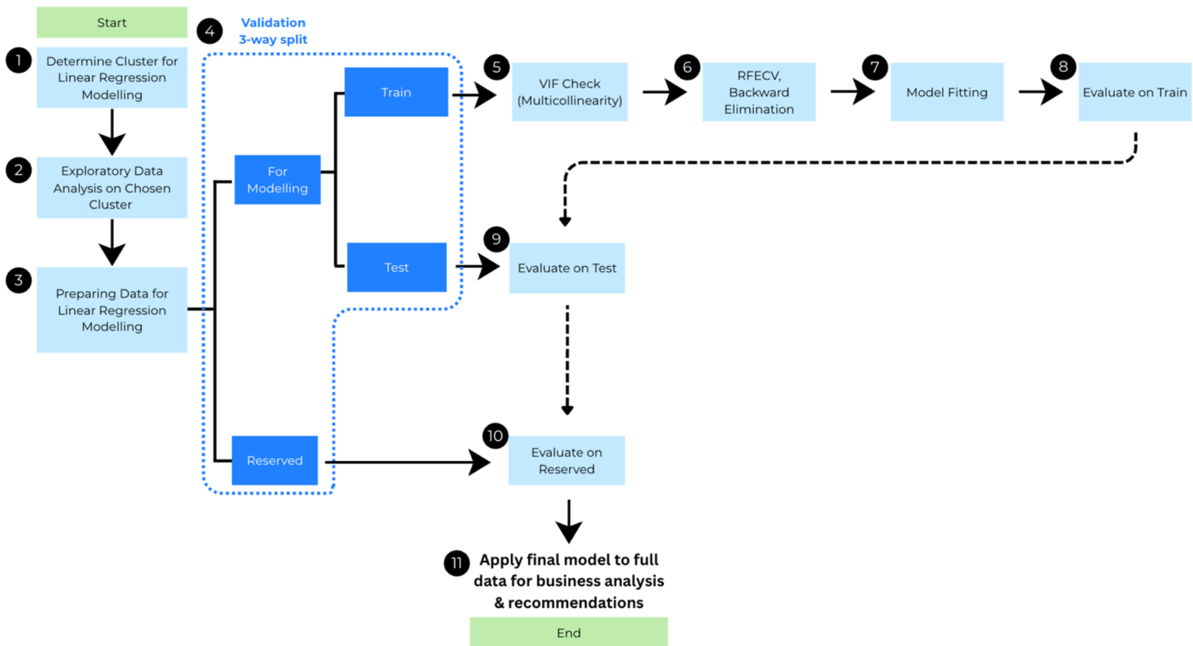


Figure C1: Overview of Linear Regression Modelling Approach Adopted

As explained in the report, Cluster 2 was selected for the linear regression modelling.

C3.1 Exploratory Data Analysis and Data Preparation – Cluster 2

Since Cluster 2 has been identified, the existing 'Cluster' column served no additional purpose and was removed. Additionally, since the information in the 'Area' column was a subset of the 'District' column, the former was therefore removed to reduce redundancy in the eventual model.

Summary statistics of Cluster 2 was generated to understand the parameters of chosen cluster data and whether there are any data quality issues which warrant further investigations:

	Bedrooms	Sq.Mt	Floor	Outer	Elevator	Penthouse	Cottage	Duplex	Semidetached	Studio	Rent
count	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00	1741.00
mean	2.36	115.10	3.63	0.88	0.89	0.09	0.00	0.00	0.00	0.00	1848.45
std	1.15	73.13	3.03	0.32	0.31	0.29	0.02	0.00	0.00	0.00	1283.65
min	0.00	20.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	450.00
25%	2.00	69.00	2.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	990.00
50%	2.00	90.00	3.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	1400.00
75%	3.00	138.00	5.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	2400.00
max	8.00	500.00	29.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	16000.00

Figure C2: Summary Statistics of Cluster 2: Mid-Sized Apartments

Exploratory analysis was also conducted on the target variable – Rent:

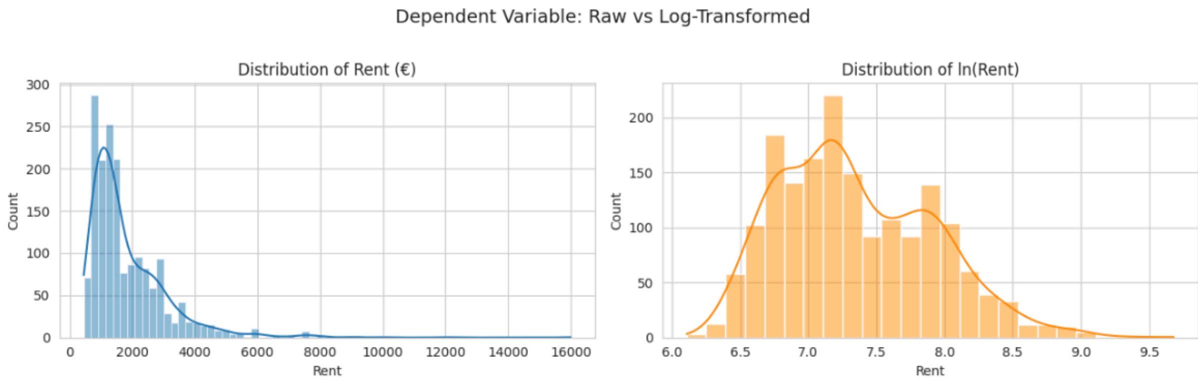


Figure C3: Distribution of Rent

As illustrated in the left histogram in Figure C3 above, rent prices were right-skewed, with the presence of high-valued apartments (e.g. €16,000). Applying a logarithmic (log) transformation stabilised the overall distribution towards a more normal distribution, as shown in the histogram chart on the right.

Bar graphs of binary features were also plotted to understand the split:

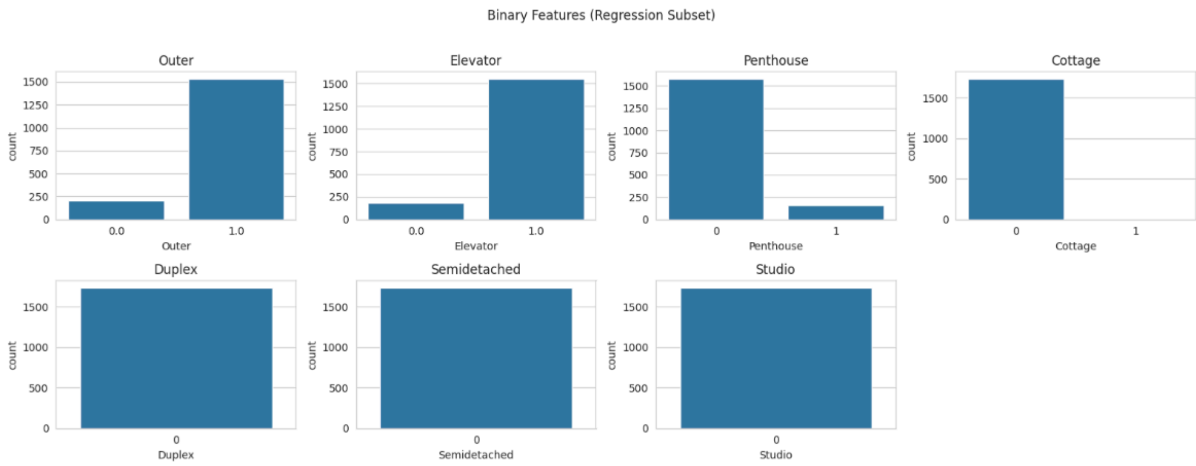


Figure C4: Bar Charts of Binary Features

Figure C4 evidently shows that the 'Duplex', 'Semidetached' and 'Studio' variables only consisted of a single type, which aligned with the profiling characteristics and labelling of Cluster 2 as Mid-Size Apartments. It is important to note that since these variables only contained a single type, they are thus considered constants and will be removed prior to modelling.

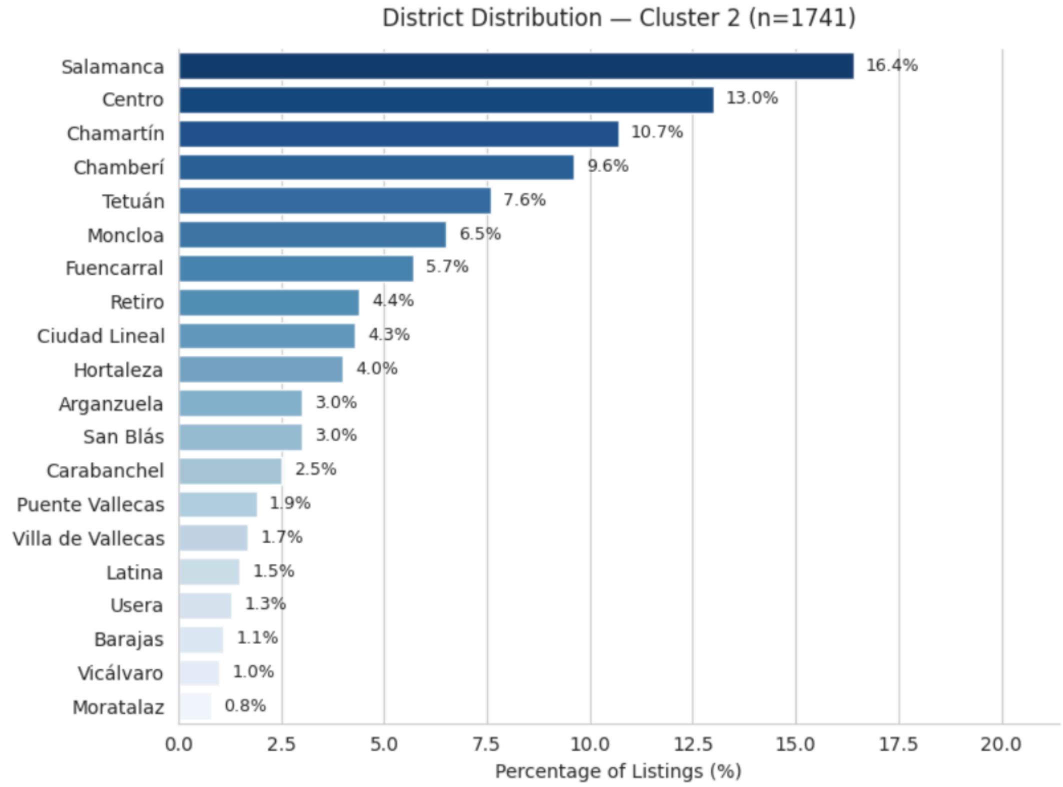


Figure C5: Distribution of Districts in Cluster 2

Most properties listed in Cluster 2 are concentrated in Salamanca (16.4%) and Centro (13.0%), as illustrated in Figure C5.

Before proceeding with the validation, one-hot encoding was performed on categorical variables (District) to facilitate analysis of their respective effects in the linear regression model.

C3.2 Validation

Prior to modelling, it was critical that the dataset is first split for validation purposes. In total, Cluster 2 data was split into three:

1. Modelling Dataset (Model) – 90% of Total
    A. Training Dataset (Train) – 70% of 90% = 63% of Total
    B. Testing Dataset (Test) – 30% of 90% = 27% of Total
2. Reserved holdout Dataset (Reserved) – 10% of Total

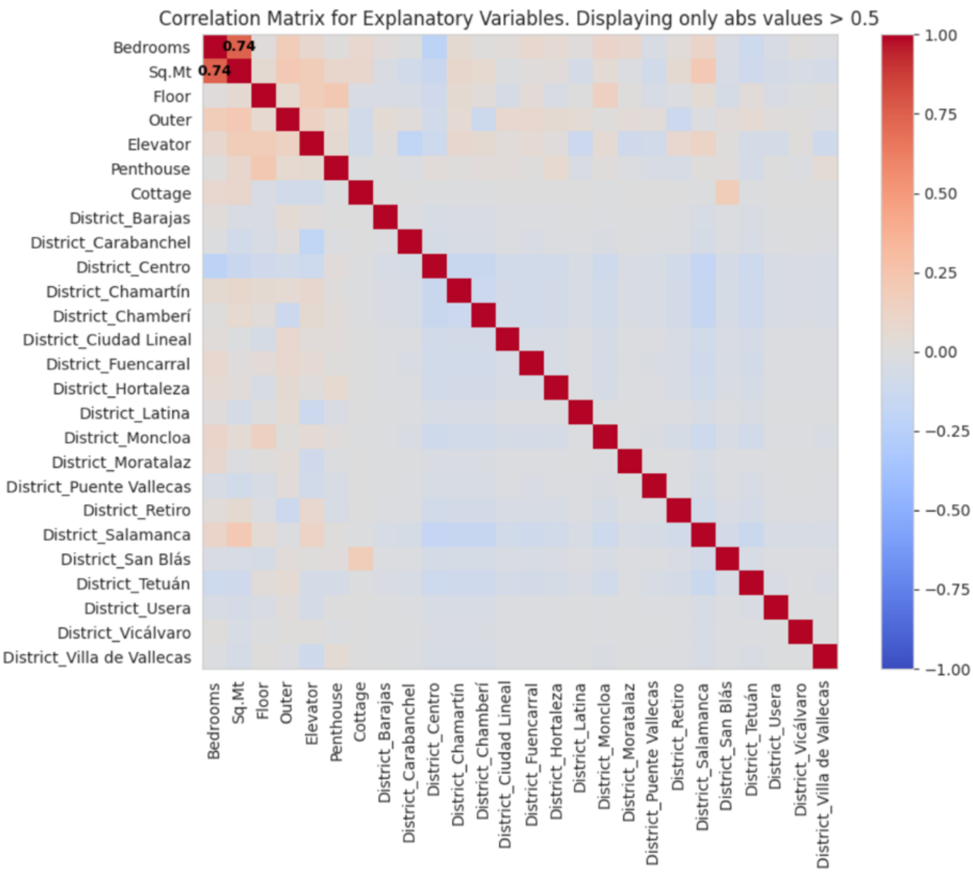
This structure allowed for model estimation on the training data, performance tuning on the test data, and final out-of-sample validation on the reserved dataset.

As identified in Part C3.1, binary variables exhibiting no within-cluster variations ('Duplex', 'Semidetached' and 'Studio') were considered as constant variables. These variables were removed since they do not command any explanatory power and keeping them may instead cause other spillover problems such as multicollinearity, thereby affecting the prediction and accuracy of the eventual regression model.

The process of building and finalising a regression model began with the Train data. A systematic approach to validate the model is as follows: Train Data → Test Data → Reserved Data.

C3.3 Multicollinearity Check

Beginning with the Train data, redundancy analysis and multicollinearity check were performed. Having redundant variables in the regression model will affect the interpretability of each explanatory variable's impact to the rent prices, therefore impacting the overall model's reliability and predictive accuracy.



Variance Inflation Factors:

	Variable	VIF
0	Bedrooms	12.88
4	Elevator	10.33
1	Sq.Mt	9.52
3	Outer	9.37
20	District_Salamanca	3.26
2	Floor	2.78
10	District_Chamartín	2.60
11	District_Chamberí	2.35
9	District_Centro	2.26
16	District_Moncloa	2.05
22	District_Tetuán	1.98
13	District_Fuencarral	1.88
12	District_Ciudad Lineal	1.64
14	District_Hortaleza	1.61
19	District_Retiro	1.57
21	District_San Blás	1.40
18	District_Puente Vallecas	1.21
8	District_Carabanchel	1.21
5	Penthouse	1.19
15	District_Latina	1.19
7	District_Barajas	1.18
25	District_Villa de Vallecas	1.16
17	District_Moratalaz	1.16
23	District_Usera	1.14
24	District_Vicálvaro	1.13
6	Cottage	1.07

Figure C6: Correlation Matrix/ Heatmap and Variance Inflation Factor (VIF) results

Based on the correlation heatmap on the left of Figure C6 above, 'Bedrooms' and 'Sq.Mt 'had the strongest correlation of 0.74 amongst all possible correlation pairs. The VIF results on the right of Figure C6 revealed that 'Bedrooms' and 'Elevator' were the only two variables that exceeded the general threshold of VIF > 10.

The higher VIF for 'Bedrooms' was consistent with the strong correlation with 'Sq.Mt', while the high VIF for 'Elevator' could be due to overlaps with 'Floor,' given that taller buildings are more than likely to have elevators. As a result, these two variables were removed to mitigate multicollinearity and improve coefficient stability.

C3.4 Feature Selection and Model Fitting

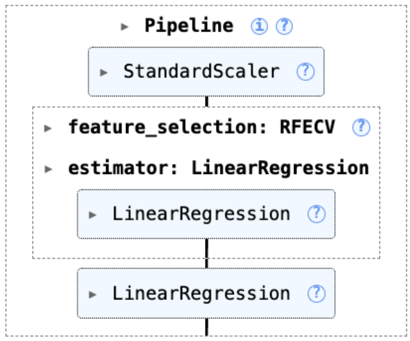


Figure C7: Modelling Pipeline

	Feature	Ranking	Selected
0	Sq.Mt	1	True
21	District_Usera	1	True
20	District_Tetuán	1	True
19	District_San Blás	1	True
18	District_Salamanca	1	True
17	District_Retiro	1	True
16	District_Puente Vallecas	1	True
15	District_Moratalaz	1	True
14	District_Moncloa	1	True
13	District_Latina	1	True
12	District_Hortaleza	1	True
11	District_Fuencarral	1	True
10	District_Ciudad Lineal	1	True
9	District_Chamberí	1	True
8	District_Chamartín	1	True
7	District_Centro	1	True
6	District_Carabanchel	1	True
5	District_Barajas	1	True
4	Cottage	1	True
3	Penthouse	1	True
2	Outer	1	True
1	Floor	1	True
22	District_Vicálvaro	1	True
23	District_Villa de Vallecas	1	True

Figure C8: Results of Recursive Feature Elimination with Cross-Validation (RFECV)

As seen in Figure C7, an overall pipeline was constructed, where feature scaling was first applied to allow for like-to-like comparison. Recursive Feature Elimination with Cross-Validation (RFECV) was performed next to identify the most predictive variables, using a 5-fold cross-validation repeated 3 times and using the negative mean squared error as the scoring metric. Results of RFECV indicated that all variables contributed to improve the model's predictive performance (Figure C8) and were therefore retained. This model resulted in a R-squared value of 0.713, indicating that 71.3% of rent prices variations were captured and explained.

```
Removing 'District_Hortaleza' – p-value: 0.5965
Removing 'District_Usera' – p-value: 0.5394
Removing 'District_Ciudad Lineal' – p-value: 0.6103
Removing 'District_Barajas' – p-value: 0.5014
Removing 'District_Puente Vallecas' – p-value: 0.5094
Removing 'District_Latina' – p-value: 0.5082
Removing 'District_San Blás' – p-value: 0.5431
Removing 'District_Villa de Vallecas' – p-value: 0.5456
Removing 'District_Carabanchel' – p-value: 0.5565
Removing 'District_Vicálvaro' – p-value: 0.3425
Removing 'District_Moratalaz' – p-value: 0.3293
Removing 'District_Fuencarral' – p-value: 0.3697
Removing 'Penthouse' – p-value: 0.1347
Removing 'Outer' – p-value: 0.0835

==============================================================================
OLS MODEL (Level)
==============================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Rent   R-squared:                       0.710
Model:                            OLS   Adj. R-squared:                  0.707
Method:                 Least Squares   F-statistic:                     265.6
Date:                Fri, 27 Feb 2026   Prob (F-statistic):          2.16e-283
Time:                        02:24:10   Log-Likelihood:                -8739.1
No. Observations:                1096   AIC:                         1.750e+04
Df Residuals:                    1085   BIC:                         1.756e+04
Df Model:                          10
Covariance Type:            nonrobust
======================================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------
const               -249.1322     53.808     -4.630      0.000    -354.712    -143.552
Sq.Mt                 13.8360      0.315     43.863      0.000      13.217      14.455
Floor                 27.6463      7.210      3.835      0.000      13.500      41.793
Cottage            -2193.4521    712.075     -3.080      0.002   -3590.652    -796.252
District_Centro      785.4804     70.604     11.125      0.000     646.944     924.016
District_Chamartín   446.3164     76.259      5.853      0.000     296.684     595.949
District_Chamberí    533.6141     76.612      6.965      0.000     383.290     683.938
District_Moncloa     408.6151     93.386      4.376      0.000     225.377     591.853
District_Retiro      614.6304    110.620      5.556      0.000     397.577     831.683
District_Salamanca   881.1650     68.397     12.883      0.000     746.960    1015.370
District_Tetuán      341.7566     84.436      4.048      0.000     176.080     507.433
==============================================================================
Omnibus:                     1119.778   Durbin-Watson:                   2.038
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           120873.290
Skew:                           4.551   Prob(JB):                         0.00
Kurtosis:                      53.636   Cond. No.                     4.49e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.49e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Figure C9: Backward elimination and Statistical Results

Backward elimination was subsequently applied at 95% confidence level, where statistically insignificant variables were removed. As seen in Figure C9, 14 variables had p-value > 0.05 and were therefore statistically insignificant and removed. The resulting model produced a R-squared of 0.710 – the negligible reduction of 0.003 affirmed that the variables removed had minimal explanatory power.

C3.5 Prediction and Evaluation: Level-Level Model on Train Dataset

Level-Level model was first performed on Train data, yielding the following results in Figure C10:

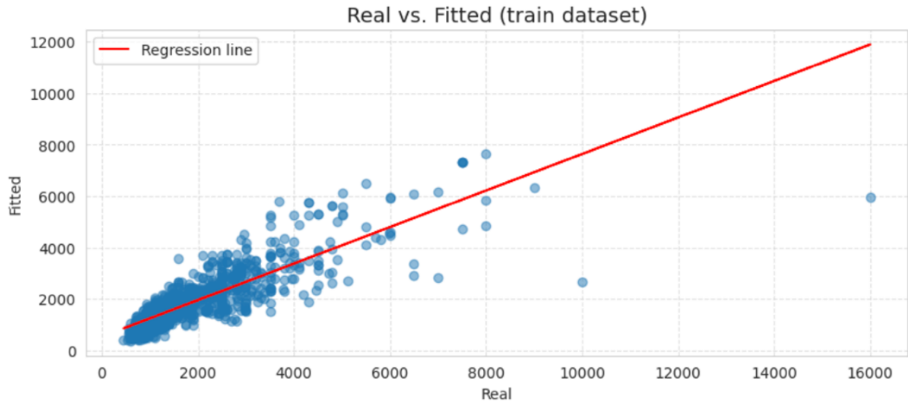


Figure C10: Level-Level Linear Regression Model (Train)

Residual diagnostics was then conducted to evaluate the model thoroughly:

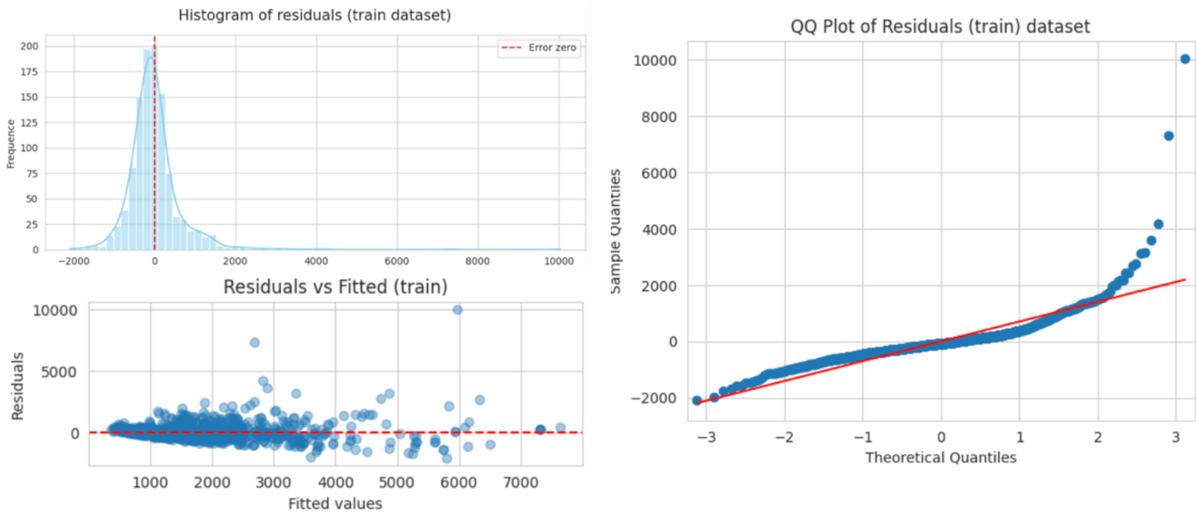


Figure C11: Residual Diagnostic on Level-Level Linear Regression Model (Train)

From Figure C11, it is evident that heteroskedasticity existed and residuals had a slight non-normal distribution. Additionally, given that 'Rent' is right-skewed *(Figure C3)*, a log transformation was explored *(see Part C3.6 below)*.

The following Predictive Performance metrics were calculated as follow:

Mean Absolute Error (MAE)	$MAE = \frac{1}{n}\sum \lvert y_i - \hat{y}_i \rvert$
Mean Squared Error (MSE)	$MSE = \frac{1}{n}\sum (y_i - \hat{y}_i)^2$
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2}$
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{100}{n}\sum \left\lvert \frac{y_i - \hat{y}_i}{y_i} \right\rvert$
R-squared	$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

where y_i= actual rent, $\hat{y}_i$= predicted rent, n= number of observations

	train
MAE	417.74
MSE	493569.49
RMSE	702.55
MAPE %	22.95
R-squared	0.7100

Figure C12: Predictive Performance Metrics: Level-Level (Train)

The Level-Level model yielded a R-squared of 0.71, indicating that 71% of variance is explained in the model.

C3.6 Prediction and Evaluation: Log-Level Model on Train Dataset

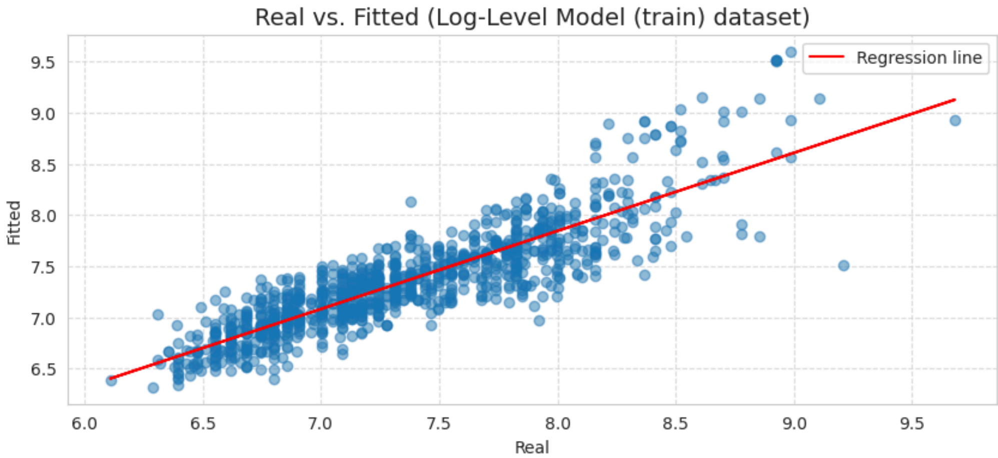


Figure C13: Log-Level Linear Regression Model (Train)

Log-Level specification was performed to address residual non-normality and heteroskedasticity observed in C3.5 above.

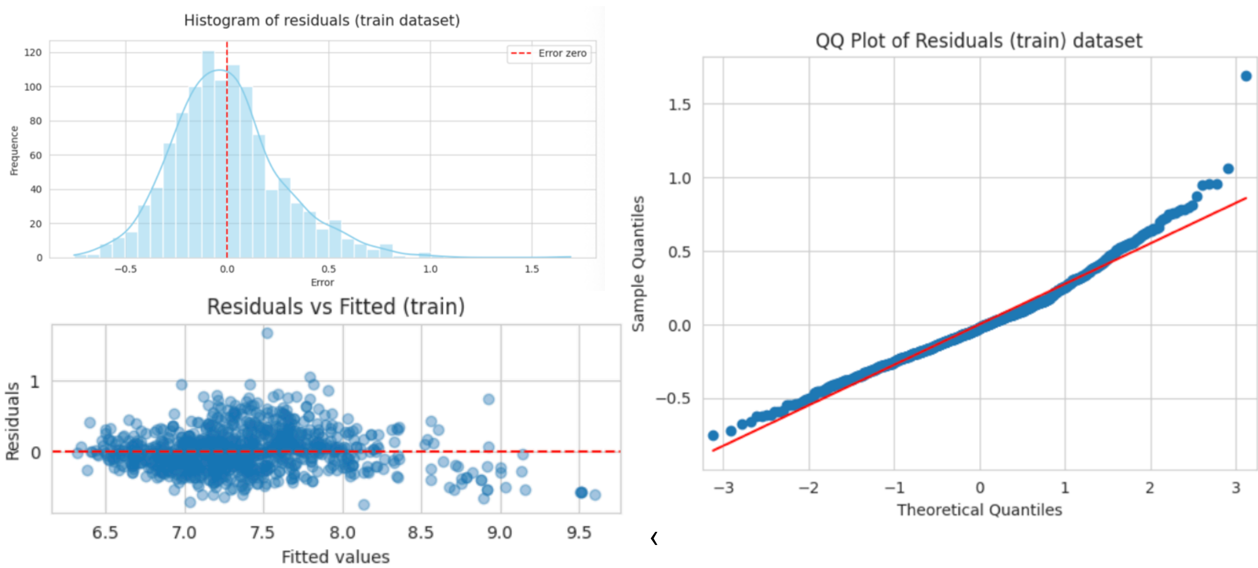


Figure C14: Residual Diagnostic on Level-Level Linear Regression Model (Train)

As expected, log transformation improved residual symmetry and reduced variance dispersion, as illustrated in Figure C14.

	Level-Level, train	Log-Level, train
MAE	417.74	448.49
MSE	493569.49	752893.27
RMSE	702.55	867.69
MAPE %	22.95	20.96
R-squared	0.7100	0.5576

Figure C15: Predictive Performance Metrics: Level-Level (Train) vs Log-Level (Train)

However, evaluation metrics showed that Level-Level model fell short of Log-Level model – reduced R-squared and increased RMSE. This indicated a deterioration in explanatory and predictive performance when measured in the original rent prices in Euros.

From a technical modelling perspective, reducing errors is one of the key focuses when determining a model suitability and performance. Ultimately, while the log transformation improved residuals behaviour, it optimises for proportional fit rather than the absolute price accuracy. Since the objective is to minimise pricing errors in Euros, the Level-Level model provided superior out-of-sample interpretability and performance.

Furthermore, right-skewness and mild heteroskedasticity are common features of any rental markets in the world due to the presence of premium properties. Considering this, any slight presence of heteroskedasticity in residuals are not totally out of the ordinary and as a result, will not materially compromise the model's predictive performance. The Level-Level model was therefore the selected model to proceed with.

C3.7 Prediction and Evaluation: Level-Level Model on Test Dataset

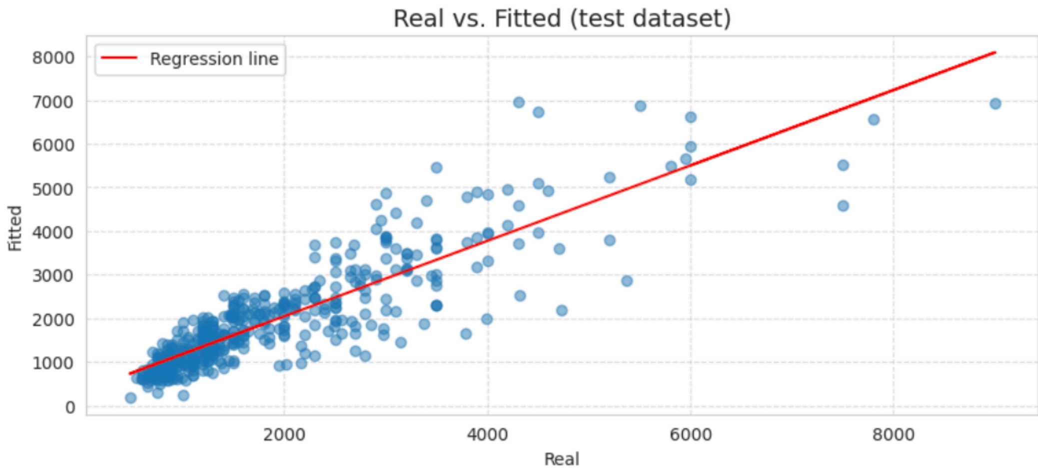


Figure C16: Level-Level Linear Regression Model (Test)

The selected Level-Level model is then applied on Test data, illustrated in Figure C16.

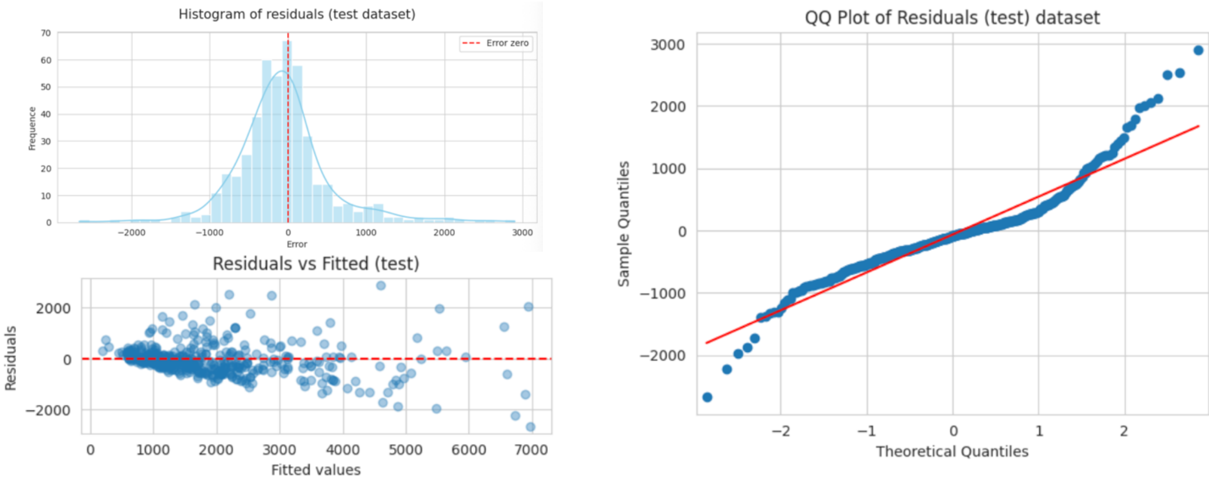


Figure C17: Residual Diagnostic on Level-Level Linear Regression Model (Test)

Residual diagnostics displayed similar patterns *(Figure C17)* to that of the Train data, with moderate right-tail dispersion at higher rent levels.

	Level-Level, train	Level-Level, test
MAE	417.74	418.34
MSE	493569.49	374554.80
RMSE	702.55	612.01
MAPE %	22.95	23.13
R-squared	0.7100	0.7490

Figure C18: Predictive Performance Metrics: Level-Level (Train) vs Level-Level (Test)

The similarity in predictive performance (Figure C18) between the Train and Test data indicated model stability and absence of overfitting. In fact, the Test data showed slight improvement in R-squared and RMSE.

C3.8 Prediction and Evaluation: Level-Level Model on Reserved Dataset

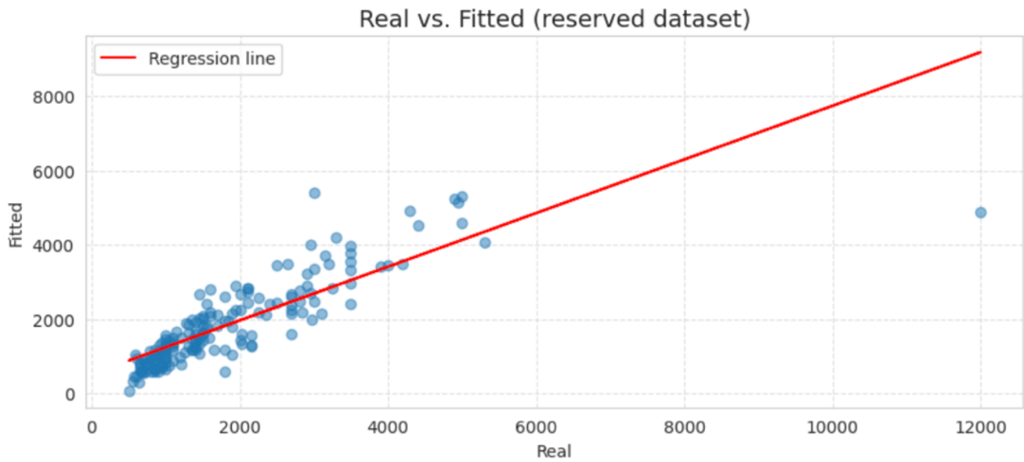


Figure C19: Level-Level Linear Regression Model (Reserved)

Finally, the Level-Level model was evaluated on the Reserved holdout data *(Figure C19)* to confirm its out-of-sample robustness.

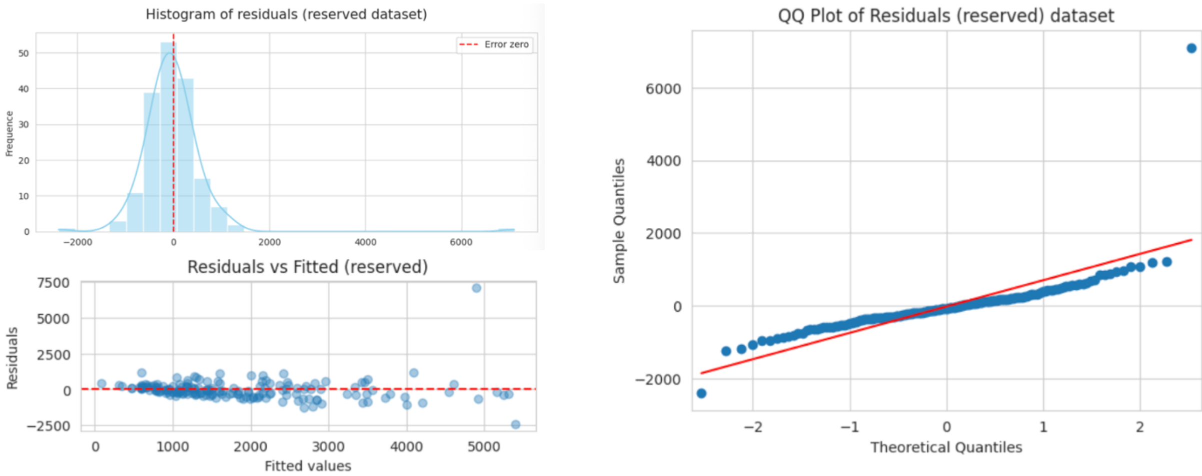


Figure C20: Residual Diagnostic on Level-Level Linear Regression Model (Reserved)

Similar to that of Part C3.7, the model performance on the Reserved dataset showed similar residual performance. The QQ plot in Figure C20 showed approximate normality, with slight deviation in the upper tail – likely due to the presence of high-value rental properties. This pattern was aligned with the general right-skewed rent price distributions and as such will not affect the stability of the model.

	Level-Level, train	Level-Level, test	Level-Level, reserved	Test vs Train	Reserved vs Train
MAE	417.74	418.34	404.18	0.61	-13.56
MSE	493569.49	374554.80	525844.51	-119014.70	32275.01
RMSE	702.55	612.01	725.15	-90.54	22.61
MAPE %	22.95	23.13	23.22	0.18	0.27
R-squared	0.7100	0.7490	0.6935	0.0390	-0.0165

Figure C21: Summary of Predictive Performance across Train, Test and Reserved Datasets

Rounding up the evaluation of the model, the Level-Level model on Reserved dataset yielded a R-squared of 0.6935 and an absolute difference of only ~0.02. Likewise, the differences in predictive performance metrics (Figure C21) were minimal and consistent with expected sampling variability. Overall, the chosen Level-Level model demonstrated stability in performance across dataset and no clear signs of overfitting.

TECHNICAL ANNEX D – PRICING RECOMMENDATIONS

The validated regression model was then used to analyse rent prices – it is important to highlight that the model was not retrained, but was simply applied to predict prices and compare against actual listed prices to identify pricing gaps and opportunities across the rental market in Madrid.

Level-Level Regression Model was applied on Cluster 2 data to derive the predicted rent prices per property listing. Pricing gaps were next computed using the following:

$$Pricing\ Gap = Actual\ Rent - Predicted\ Rent$$

$$Pricing\ Gap\ \% = \frac{Pricing\ Gap}{Predicted\ Rent} \times 100$$

The Pricing Gap % was then used to identify 3 main buckets: (1) Underpriced, (2) Fairly Priced and (3) Overpriced. Underpriced properties were defined as properties priced at least -20% below predicted prices, while Overpriced properties are defined as properties priced at least +20% above predicted prices.

```
Actual vs Predicted Prices
Actual vs Predicted
Fairly Priced    935
Underpriced      437
Overpriced       369
Name: count, dtype: int64

Average pricing gap: 1.97%
Median pricing gap: -5.69%
Std of pricing gap: 35.20%
```

Figure D1: Summary of Comparison of Rent Prices – Actual vs Predicted

From Figure D1, of the 1,741 properties in Cluster 2, 935 (53.7%) were fairly priced, 437 (25.1%) were underpriced and 369 (21.2%) were overpriced.

A summary of pricing gaps by districts was created, applying colour scale to denote the range of deviation, with green colour used for underpriced properties and red for overpriced properties:

District	listings	avg_actual	avg_predicted	underpriced	overpriced	underpriced proportion %	overpriced proportion %
Salamanca	286	2863.00	2861.26	87	58	30.4	20.3
Centro	227	1779.00	1873.91	81	34	35.7	15.0
Chamartín	187	2093.00	2114.20	44	27	23.5	14.4
Chamberí	167	2178.00	2178.33	37	37	22.2	22.2
Tetuán	133	1449.00	1442.72	30	22	22.6	16.5
Moncloa	113	2003.00	2058.93	39	25	34.5	22.1
Fuencarral	99	1407.00	1509.32	14	19	14.1	19.2
Retiro	76	2315.00	2330.66	28	12	36.8	15.8
Ciudad Lineal	74	1444.00	1380.82	6	27	8.1	36.5
Hortaleza	69	1551.00	1459.64	8	18	11.6	26.1
San Blás	53	1062.00	1091.18	6	16	11.3	30.2
Arganzuela	53	1228.00	1045.89	4	32	7.5	60.4
Carabanchel	43	745.00	781.74	12	10	27.9	23.3
Puente Vallecas	33	746.00	718.49	6	13	18.2	39.4
Villa de Vallecas	29	857.00	913.47	8	4	27.6	13.8
Latina	26	890.00	966.21	8	4	30.8	15.4
Usera	22	821.00	844.47	6	5	27.3	22.7
Barajas	19	1086.00	1178.09	3	3	15.8	15.8
Vicálvaro	18	784.00	911.47	5	1	27.8	5.6
Moratalaz	14	949.00	1159.08	5	2	35.7	14.3

Figure D2: Summary of Pricing Gaps by Districts

The top 10 underpriced and overpriced properties are further broken down here to identify clear opportunities:

	District	Sq.Mt	Bedrooms	Floor	Rent	predicted_rent	pricing_gap_pct	Actual vs Predicted
808	Salamanca	200.00	2.00	7.00	1600.00	3592.75	-55.47	Underpriced
677	Salamanca	85.00	2.00	-0.50	925.00	1794.27	-48.45	Underpriced
354	Latina	150.00	1.00	3.00	1000.00	1909.20	-47.62	Underpriced
1154	Centro	50.00	2.00	5.00	730.00	1366.38	-46.57	Underpriced
2070	Vicálvaro	98.00	3.00	7.00	700.00	1300.32	-46.17	Underpriced
703	Salamanca	136.00	4.00	6.00	1450.00	2679.60	-45.89	Underpriced
698	Salamanca	95.00	2.00	3.00	1100.00	2029.39	-45.80	Underpriced
709	Salamanca	53.00	1.00	7.00	850.00	1558.86	-45.47	Underpriced
1268	Centro	80.00	2.00	2.00	940.00	1698.52	-44.66	Underpriced
505	Moncloa	102.00	3.00	0.00	870.00	1570.75	-44.61	Underpriced

Figure D3: Top 10 Underpriced Properties

	District	Sq.Mt	Bedrooms	Floor	Rent	predicted_rent	pricing_gap_pct	Actual vs Predicted
989	Arganzuela	25.00	1.00	-0.50	500.00	82.94	502.82	Overpriced
1020	Arganzuela	30.00	1.00	3.00	999.00	248.89	301.39	Overpriced
1926	Tetuán	185.00	2.00	1.00	10000.00	2679.92	273.15	Overpriced
92	Ciudad Lineal	55.00	1.00	3.00	1800.00	594.78	202.63	Overpriced
1188	Centro	390.00	4.00	1.00	16000.00	5960.02	168.46	Overpriced
1004	Arganzuela	35.00	1.00	2.00	750.00	290.42	158.25	Overpriced
427	Moncloa	56.00	1.00	7.00	2800.00	1127.82	148.27	Overpriced
581	Puente Vallecas	30.00	1.00	1.00	480.00	193.59	147.94	Overpriced
902	Salamanca	155.00	3.00	2.00	7000.00	2831.90	147.18	Overpriced
761	Salamanca	300.00	3.00	4.00	12000.00	4893.40	145.23	Overpriced

Figure D4: Top 10 Overpriced Properties